

# Subtract mixed numbers

Use the scaffold, show each step and ask for an adult check when marked.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Read the model: Calculate  $5 \frac{1}{4} - 2 \frac{2}{3}$ . Copy or complete the first step.

2. A jug contains  $3 \frac{1}{2}$  litres.  $1 \frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?

3. Calculate  $\frac{3}{4} + \frac{5}{6}$ .

4. Miro says: Miro adds or subtracts both the numerators and denominators. Is this correct? Explain.

# Teacher answers and checking prompts

1. Read the model: Calculate  $5\frac{1}{4} - 2\frac{2}{3}$ . Copy or complete the first step.

**Answer**  $2\frac{7}{12}$ .

2. A jug contains  $3\frac{1}{2}$  litres.  $1\frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?

**Answer**  $2\frac{5}{12}$  litres.

3. Calculate  $\frac{3}{4} + \frac{5}{6}$ .

**Answer**  $1\frac{7}{12}$ .

4. Miro says: Miro adds or subtracts both the numerators and denominators. Is this correct? Explain.

**Answer** Use equivalent fractions with a common denominator, then operate on the numerators only.

# Subtract mixed numbers

Work independently. Show a complete method and check at least one answer.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Calculate  $5\frac{1}{4} - 2\frac{2}{3}$ .

6. Check this result using an inverse, estimate or known fact: Calculate  $\frac{3}{4} + \frac{5}{6}$ .

2. A jug contains  $3\frac{1}{2}$  litres.  $1\frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?

7. Prove or explain your reasoning: Calculate  $\frac{7}{8} - \frac{5}{12}$ .

3. Calculate  $\frac{3}{4} + \frac{5}{6}$ .

8. Identify and correct this misconception: Miro adds or subtracts both the numerators and denominators.

4. Solve this independently and show every stage: Calculate  $5\frac{1}{4} - 2\frac{2}{3}$ .

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: A jug contains  $3\frac{1}{2}$  litres.  $1\frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?

10. Write and solve a similar question to: Calculate  $\frac{3}{4} + \frac{5}{6}$ .

# Teacher answers and checking prompts

1. Calculate  $5 \frac{1}{4} - 2 \frac{2}{3}$ .  
**Answer**  $2 \frac{7}{12}$ .
2. A jug contains  $3 \frac{1}{2}$  litres.  $1 \frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added.  
How much is in the jug?  
**Answer**  $2 \frac{5}{12}$  litres.
3. Calculate  $\frac{3}{4} + \frac{5}{6}$ .  
**Answer**  $1 \frac{7}{12}$ .
4. Solve this independently and show every stage: Calculate  $5 \frac{1}{4} - 2 \frac{2}{3}$ .  
**Answer**  $2 \frac{7}{12}$ .
5. Use a second method or representation for: A jug contains  $3 \frac{1}{2}$  litres.  $1 \frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?  
**Answer** Expected result:  $2 \frac{5}{12}$  litres. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Calculate  $\frac{3}{4} + \frac{5}{6}$ .  
**Answer** Expected result:  $1 \frac{7}{12}$ . A valid check must be shown.
7. Prove or explain your reasoning: Calculate  $\frac{7}{8} - \frac{5}{12}$ .  
**Answer**  $\frac{11}{24}$ . The explanation must show why the method works.
8. Identify and correct this misconception: Miro adds or subtracts both the numerators and denominators.  
**Answer** Use equivalent fractions with a common denominator, then operate on the numerators only.
9. Write a complete sentence explaining how you checked one answer.  
**Answer** Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Calculate  $\frac{3}{4} + \frac{5}{6}$ .  
**Answer** Answers vary. The new question must use the same mathematical structure and include a correct solution.

# Subtract mixed numbers

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: \_\_\_\_\_ Date: \_\_\_\_\_

1. Prove or explain your reasoning: Calculate  $7/8 - 5/12$ .

6. Create a new example that tests the same idea as: Calculate  $3/4 + 5/6$ .

2. Prove the answer to this question, rather than only calculating it: Calculate  $5 \frac{1}{4} - 2 \frac{2}{3}$ .

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: A jug contains  $3 \frac{1}{2}$  litres.  $1 \frac{3}{4}$  litres are used and  $2/3$  litre is added. How much is in the jug?

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question:  $1 \frac{7}{12}$ .

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro adds or subtracts both the numerators and denominators.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

# Teacher answers and checking prompts

1. Prove or explain your reasoning: Calculate  $7/8 - 5/12$ .  
**Answer**  $11/24$ . The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Calculate  $5\frac{1}{4} - 2\frac{2}{3}$ .  
**Answer** Expected result:  $2\frac{7}{12}$ . Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: A jug contains  $3\frac{1}{2}$  litres.  $1\frac{3}{4}$  litres are used and  $\frac{2}{3}$  litre is added. How much is in the jug?  
**Answer** Expected result:  $2\frac{5}{12}$  litres. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question:  $1\frac{7}{12}$ .  
**Answer** One valid reconstruction is: Calculate  $3/4 + 5/6$ . Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro adds or subtracts both the numerators and denominators.  
**Answer** Use equivalent fractions with a common denominator, then operate on the numerators only.
6. Create a new example that tests the same idea as: Calculate  $3/4 + 5/6$ .  
**Answer** Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.  
**Answer** Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.  
**Answer** Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.  
**Answer** Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.  
**Answer** Answers vary. The explanation must distinguish the mathematical structure from the changed value.